

Objective

Hands-on Software Architect and Principal Software Engineer with deep expertise in distributed systems, data architecture, and stream processing. Brings strong experience in Java, Apache Flink, and cloud-native platforms, with a track record of designing and delivering scalable, real-time data systems. Established as a leader, mentor, and active contributor to engineering communities. Experience spans ad tech, AI platforms, and large-scale data systems, with a consistent focus on architecture, concurrency, high-performance distributed systems, and data-intensive applications.

Seeking roles as Software Architect, Data Architect, or Principal Engineer (Full-Time or Corp-to-Corp). Open to remote.

Experience

Disney Streaming, Inc.

Principal Software Engineer

Nov 2025 – Present

Served on the Software Architecture team for the Disney Ad Server platform enabling monetization of Disney+ and Hulu.

- Contributed to architecture committees defining platform-wide architecture standards for domain modeling, CI/CD (trunk-based development), feature flags (LaunchDarkly), and AI context engineering across multiple cross-functional engineering teams.
- Delivered UI Standards Architecture using Attribute-Driven Design (ADD), leveraging AI-assisted development (Cursor, GitHub Copilot).
- Designed a data lake architecture and delivered a CDC proof of concept from MySQL to Apache Iceberg, enabling schema evolution and time travel.
- Built an end-to-end CDC pipeline (Flink CDC, Debezium, Paimon, Spark, S3) deployed via Kubernetes and Terraform, demonstrating production-ready patterns.

GushR AI, Inc.

Software Architect (Independent / After-hours)

October 2024 – Present

Software Architect responsible for transforming AI proof of concepts into a production-ready, multi-tenant Platform-as-a-Service offering for "AI-Powered Methane Monitoring for Real-Time Emission Intelligence." Led end-to-end architecture and hands-on implementation across data, infrastructure, and application layers, incorporating AI-assisted development practices.

- Architected and built a multi-tenant PaaS platform, owning system design, implementation, and deployment strategy.
 - Designed and implemented a video-based AI inference pipeline supporting both batch and real-time processing using a kappa architecture.
 - Built streaming and data processing systems with Apache Flink, integrating video ingestion with speed layers (InfluxDB, PostgreSQL) and long-term storage (Apache Iceberg on S3).
 - Developed data orchestration and control-plane APIs using Spring Boot.
 - Implemented batch AI inference workflows using Celery for scalable asynchronous processing.
 - Provisioned and managed cloud-native infrastructure using Kubernetes, Terraform, Minikube, and GitHub, supporting deployments to AWS and LocalStack.
 - Built UI demos and prototype interfaces using Next.js, Tailwind CSS, and TypeScript to support productization and stakeholder validation.
 - Leveraged AI-assisted development (context engineering tools) to accelerate architecture design and implementation.
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Systech Solutions, Inc.

Software Architect

Jun 2024- Sep 2025

Remote

Led a team of software architects in delivering the company's next-generation cloud-based track-and-trace solutions.

- Led the development and implementation of standardized operating procedures for software architecture, utilizing Attribute-Driven Design 3.0 and adapting it for agile development.
- Led a five-person architecture team, driving cohesive design decisions aligned with business goals and technical objectives.
- Delivered a comprehensive Reference Architecture for Spring Boot-based microservices, optimized for PostgreSQL and MongoDB databases.
- Coordinated the iterative design and development of software architecture across five parallel development tracks, consistently meeting aggressive project timelines.

Beyond Limits, Inc.

Software Architect

Jan 2020- Dec 2023

Glendale, CA

Software Architect with hands-on engineering responsibility for the architecture, data strategy, and implementation of multiple hybrid-AI SaaS products.

- Designed cloud-agnostic, but deployed to AWS, real-time IoT data platforms using Apache Flink, Spring Boot, PostgreSQL, InfluxDB, S3, and React.
- Architected real-time scalable IoT data ingestion platforms using kappa architecture, Apache Flink, with long-term storage in S3/Parquet and speed layers in InfluxDB and PostgreSQL.
- Built Spring Boot microservices, PostgreSQL schemas with Liquibase, and fully containerized services.
- Created an ANTLR-based expression language for real-time evaluation of time-series data within an Apache Flink Stateful Functions application.
- Delivered architecture using SEI-inspired Attribute-Driven Design.
- Mentored teams on engineering standards and best practices.

Walt Disney Direct-to Consumer & International

Sr. Software Engineer (Contractor)

Apr 2019- Jan 2020

Glendale, CA

Integrated Disney's Direct to Consumer Streaming Technology assets with the RightsLine digital rights management PaaS.

- Designed and developed a business rules engine to compute digital rights based on specified rules and airdates. Employed the Attribute-Driven Design methodology (ADD 3.0) to deliver a well-structured architecture.
 - Collaborated closely with a team of developers to maintain and enhance a critical data pipeline responsible for publishing digital rights into the RightsLine digital rights management system. Utilized a tech stack comprised of Spring Boot, GitLab, Jenkins, Kubernetes, EKS, Kotlin, Java, and PostgreSQL.
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Gamblit Gaming, LLC

Sr. Software Engineer

Nov 2015- Jan 2019

Glendale, CA

We developed software for casino gaming systems (Las Vegas). These were skill-based casino video games with touch screens, bill validators, printers, that must interface with casinos via the Slot Accounting System (SAS) protocol. My role was software architecture, hands-on development, mentoring, and technical leadership.

- Architected and developed RESTful APIs utilizing Java, JDK 1.8, and Spring Boot. Defined a well-structured layered architecture, package organization, and build processes. Provided mentorship on best practices in concurrency programming, domain-driven design, and Spring Framework. Technologies leveraged dependency injection, transaction management with AOP, JAX-RS with Jersey, JSON serialization with Jackson, JDBC, Docker, Docker Compose, Flyway database migrations, PostgreSQL, and Microsoft SQL Server.
- Engineered a streaming data statistical calculator assessing randomness with the Runs Test and Autocorrelation Test, capable of handling extensive datasets and input files exceeding 10 GB. Implemented algorithms from scratch based on academic research papers from NIST.
- Implemented high-performance, low-latency TCP/IP application layer protocols utilizing Java 8, NIO, and Netty.
- Created a Java-based command-line tool integrated with Git, facilitating the analysis of Maven projects and automated checking out of dependencies in the organization's designated build or tagging order. Leveraged Maven Aether and JGit libraries.
- Led the architecture and development of a full-stack system, employing Attribute-Driven Design methodology (ADD 3.0). Orchestrated UI development with Angular 6 and Bootstrap, RESTful API development with Python 2, Flask, and SQL Alchemy, and OAuth 2 Authorization server implementation with Python 2, Flask, Authlib, Open LDAP, and SQL Alchemy.

Loot Crate

Sr. Software Engineer

Apr 2015- Nov 2015

Los Angeles, CA

Led the transition from a monolithic Ruby on Rails application focusing on subscription and shipping services to a distributed architecture, meticulously assessing key components to address performance and scalability challenges while recommending appropriate technical solutions.

- Engineered and executed database migration scripts, enhancing the production PostgreSQL database with referential integrity, auditing functionality, and data anomaly fixes.
- Created and implemented an internal Ruby GEM named "Badger," streamlining the development of high-performing database batch programs. This significantly reduced execution time, cutting it down from 18 hours to an impressive 15 minutes when compared to an ordinary MRI Ruby application.
- Led the development of a high-performance, multi-threaded ETL data pipeline, efficiently handling 900,000 records, persisting 7 million database records, and uploading 3.5 million files totaling 73.3 GB in S3. Leveraged JRuby, Java threads, Java Concurrency, JDBC batch, Badger, and AWS for this project.

Overstock.com

Sr. Software Engineer (Contractor)

Oct 2012- Apr 2015

South Pasadena, CA

Engaged as a remote consultant on several Service Oriented Architecture (SOA) projects, specializing in the development and maintenance of RESTful Web Services, batch processing systems, and web applications.

- Collaborated in the development of tailored SOA solutions, including Mule SOA solutions, primarily focusing on RESTful APIs utilizing Spring Boot, JAX-RS, and JPA.
 - Engineered efficient, stateless client libraries for HTTP REST, SFTP, and FTP, integral to our SOA components, enhancing communication and data transfer capabilities.
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- Implemented Docker containers for streamlined integration testing, ensuring robust testing procedures.
- Architected a distributed RESTful file system utilizing Netty and Cassandra.
- Developed high-performing data pipelines designed to handle GB-sized XML payloads and batched information into a database. Utilized technologies such as Spring Boot, JAX-RS, StAX, and JDBC for optimal product performance.

SupplyFrame Media

Lead Software Engineer

Apr 2011- Sep 2012

Los Angeles, CA

Served as the Lead Developer for SupplyFrame Media's primary aggregate e-commerce platform, overseeing a dedicated development team.

- Orchestrated the integration of multiple vendors into our e-commerce platform. This project utilized technologies such as Tomcat, Struts 2, Spring Framework, and PostgreSQL within a monolithic architecture.
- Led the development and implementation of advanced full-text search capabilities utilizing Lucene, enhancing the search functionality and user experience on our e-commerce platform.
- Led the development of RESTful APIs using JAX-RS, optimizing interoperability within the platform.
- Engineered Client-Server applications and TCP/IP application layer protocols, utilizing NIO, Netty, and Protocol Buffers to ensure efficient transmission.

The Rubicon Project

Senior Software Engineer

Nov 2010-Apr 2011

Los Angeles, CA

Facilitated the technology transfer of the Fox Audience Network (FAN) Ad Server during the acquisition by the Rubicon Project. Additionally, led the development of an innovative "agile" meta-programming framework, capable of generating RESTful web APIs using SQL queries as input.

NewsCorp

Senior Software Engineer

Jun 2007-Oct 2010

Fox Audience Network

Santa Monica, CA

I was a key member of a small team responsible for developing and maintaining a high-performance, optimizing display ad server that reached its peak of 7 billion impressions per day.

- Led the development of a robust, mission-critical micro-batch data processing pipeline dedicated to aggregating all display ad impressions. Employed concurrency programming and database batch techniques, achieving an impressive rate of approximately 2000 transactions per second. Technologies utilized included CSV, Java, JDBC, and DB2.
- Contributed to the maintenance and enhancement of our Ad Server, a critical component of our system, implementing the 2nd Price Auction and real-time cookie encryption/decryption routines. Leveraged Java and Servlet for this project.
- Developed back-end web services integral to the functionality of MyAds and Publisher Network Portal products, allowing campaign management on our Ad Server. Technologies involved were Java, JAX-WS, JAXB, Spring Framework, Hibernate, and DB2.
- Directed the development of an XML file batch processing system that automated the creation of display ads. This innovative system demonstrated remarkable efficiency, processing 1 GB XML files in under a minute. The project utilized Java, DOM, StAX, and JDBC batch programming.

Senior Software Engineer

Oct 2006-Jun 2007

MySpace

Beverly Hills, CA

Supported the maintenance of a monolithic MVC web application with Social Networking and Karaoke functionalities. Primarily utilized Java, Apache Tomcat, Struts, Spring Framework, and MySQL.

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Additionally, successfully integrated advanced full-text search capabilities using Lucene, enhancing the application's search functionality.

Education	1996–1999	University of California Irvine	B.S., Information and Computer Science
	1992–1995	Pasadena City College	Associates of Arts
Skills	Java, JEE, J2EE, EJB, JDBC, JPA, Web Services, Servlet, JSP, JSF, JAX-RS, JAX-WS, REST, SOAP, OAuth 2, JSON and JAAS.		

Apache Flink, Apache Flink Stateful Function, Stream Programming, Data Architecture, Kappa Architecture, Lakehouse Architecture, Parquet, Apache Paimon, Apache Iceberg, Apache Spark, Apache Flink CDC, Flink Kubernetes Operator, Apache Kafka, Strimzi Kafka Operator, and Debezium.

AI, LLM, Context Engineering, Cursor, Claude Code, Cline, GitHub Co-Pilot, Skills, MCP Server, Ollama, Hugging Face, and Web UI.

The Spring Framework, Spring Boot, Spring MVC, Spring Security, Inversion of Control, Dependency Injection, and AOP.

Network Programming, Socket Programming, NIO, Netty, and Protocol Buffers.

SQL, RDBMS, PostgreSQL, Oracle, DB2, Microsoft SQL Server, MySQL, SQLite, H2, Liquibase, Flyway, ACID, Normalization, Third Normal Form (3NF).

NoSQL, Mongo DB Atlas, Influx DB, Elastic Search, Lucene, Phoenix, HBase, Cassandra, DynamoDB, Hadoop, and Map Reduce.

Concurrency programming, Multi-threaded programming, Thread-safety, and Java Util Concurrent.

Kubernetes, MiniKube, Containers, Docker, Docker Compose, and Terraform.

AWS, Cloud Computing, Distributed Computing, EC2, ECS, EKS, EMR, and LocalStack.

Python, Flask, SQL Alchemy, Authlib, UV, PyEnv, and Virtual Env.

Ruby, JRuby, and Ruby on Rails.

Angular, NextJS, React, TypeScript, JavaScript, ECMAScript, JQuery, Tailwind, CSS, and Bootstrap.

Linux, Open LDAP, Bash, Awk, and sometimes PERL.

Version Control, Git, GitHub, GitHub Actions, GitLab, Bitbucket, Subversion, and CVS.

Package Management, Maven, Gradle, PIP, NPM, and Bundler.

Software Architecture, Attribute Driven Design 3.0, UML, Quality Attributes, Architecture Patterns, Architecture Tactics, and Software Engineering.

Agile, Scrum, Use Cases, and Stories.

Verbal, written, and group presentations.

Affiliations

Los Angeles Java Users' Group

President

May 2004-Present

Non-Profit Organization

Los Angeles, CA

Led and managed the Los Angeles Java User's Group since 2004, overseeing all aspects of its operations. This encompassed organizing meetings, curating pertinent technical topics, securing knowledgeable presenters, ensuring suitable meeting venues, hosting and moderating meetings, fostering connections between attendees and recruiters, and maintaining the group's website.

Presentations

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- Understanding Apache Flink
 - Recognizing the Common Pitfalls in Java Concurrency Programming
 - Improving Your Relational Database Architecture
 - Delivering Quality Software with Modern Unit Testing Techniques
 - Tips in Concurrency
 - Persistence with JPA 2.1
 - Get there Faster with JRuby and JRuby on Rails
 - RESTful JSON Web Services with Jersey & Jackson
 - Java Concurrency